

A deep infrared and spectroscopic survey of dust-obscured starburst galaxies in the Apex Deep Field

Elena R. Vance,^{1*} Marcus Thorne,^{1,2} Aria Lin,² Julian K. Sterling³ and Sophia H. Rossi⁴

¹Department of Astrophysics, Meridian Institute of Advanced Study, 100 Observatory Ridge, Cambridge CB3 0HA, UK

²Center for Computational Astrophysics, Apex Observatory, 45 Science Park Way, Pasadena, CA 91125, USA

³Institute for Extra-Galactic Astronomy, Vertex University, 500 University Avenue, Munich D-80333, Germany

⁴Department of Physics and Astronomy, Synapse National Laboratory, Los Alamos, NM 87545, USA

Accepted 2026 June 12. Received 2026 June 01; in original form 2026 March 15

ABSTRACT

We present deep multi-wavelength observations of dust-obscured star-forming galaxies (DSFGs) in the Apex Deep Field (1.5 deg^2) combining submillimetre continuum mapping from the Meridian-Submm Array with high-resolution near-infrared spectroscopy from Vertex-VLT. We identify a sample of 42 high-redshift ($2.0 < z < 4.5$) DSFGs with extreme star formation rates ($\text{SFR} \sim 300\text{--}1800 \text{ M}_\odot \text{ yr}^{-1}$) and massive stellar components ($M_* \sim 10^{10.5}\text{--}10^{11.8} \text{ M}_\odot$). Kinematic modeling of the $\text{H}\alpha$ and $[\text{O III}] \lambda 5007$ emission lines reveals that 64 ± 8 per cent of the sample exhibit rotation-dominated disk dynamics with velocity dispersions $\sigma_0 = 45 \pm 12 \text{ km s}^{-1}$, whereas the remaining 36 per cent show disturbed, merger-driven kinematics. The gas-phase metallicities derived via the $N2$ and $O3N2$ calibrations indicate that these systems lie on an extended mass-metallicity relation consistent with rapid chemical enrichment fuelled by cold gas accretion streams. We find a mean gas mass fraction of $f_{\text{gas}} = 0.52 \pm 0.09$, corresponding to depletion timescales of $\tau_{\text{dep}} \approx 150 \text{ Myr}$. These results strongly suggest that high-redshift DSFGs represent the main progenitor population of massive, passively evolving elliptical galaxies observed at $z \sim 2$.

Key words: galaxies: evolution – galaxies: high-redshift – galaxies: ISM – galaxies: starburst – submillimetre: galaxies

1 INTRODUCTION

The peak epoch of cosmic star formation rate density ($z \sim 2\text{--}3$), often referred to as “cosmic noon”, is characterised by intense growth of galactic stellar mass (Madau & Dickinson 2014; Sterling et al. 2021). A substantial fraction of this star formation activity occurs within dust-obscured star-forming galaxies (DSFGs), which are characterised by infrared luminosities $L_{\text{IR}} > 10^{12} L_\odot$ and submillimetre flux densities $S_{850 \mu\text{m}} \geq 2.5 \text{ mJy}$ (Vance et al. 2020; Thorne et al. 2023). Due to heavy dust attenuation in the ultraviolet and optical regimes, tracing the physical processes governing their assembly requires long-wavelength imaging paired with high-sensitivity infrared integral field spectroscopy (Lin et al. 2022).

Early theoretical frameworks postulated that high-redshift starbursts were exclusively triggered by major gas-rich mergers (Rossi et al. 2019). However, recent high-resolution hydrodynamic simulations from the Apex-Suite project suggest that continuous cold gas streams from the cosmic web can sustain high-rate star formation within extended, gas-rich turbulent disks without destroying disk morphology (Sterling et al. 2024). Disentangling the relative contributions of disk order versus merger chaos in DSFGs is therefore central to constraining galaxy transformation mechanisms across cosmic time.

In this paper, we present results from the Meridian-Apex Deep Field Survey (MADFS). In Section 2, we detail our observational campaign and data reduction protocols. Section 3 describes the sample selection and physical parameter extraction. Kinematic structures and

gas-phase metallicities are analysed in Section 4. In Section 5, we discuss the cosmological implications of our findings, and Section 6 summarises our principal conclusions. Throughout this paper, we adopt a flat Λ CDM cosmology with $\Omega_m = 0.307$, $\Omega_\Lambda = 0.693$, and $H_0 = 67.7 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Planck Collaboration 2020).

2 OBSERVATIONS AND DATA REDUCTION

2.1 Submillimetre Mapping with Meridian-SMA

Observations of the 1.5 deg^2 Apex Deep Field were conducted using the Meridian-Submm Array (MSMA) located at the Apex High Observatory between October 2024 and February 2025. The array was configured with 12 antenna elements providing baselines ranging from 15 m to 510 m. Continuum mapping at $850 \mu\text{m}$ (345 GHz) achieved a median synthesised beam full-width at half-maximum (FWHM) of $0.45'' \times 0.38''$.

Data reduction was executed using the standard APEX-REDUCE pipeline version 4.2. Raw visibilities were calibrated for atmospheric opacity using sky dip scans, while quasar J0521+1638 served as the primary phase calibrator. The final mosaics reached a uniform rms noise level of $\sigma_{850} = 0.18 \text{ mJy beam}^{-1}$.

2.2 Near-Infrared Spectroscopic Follow-up

Spectroscopic target confirmation and kinematic mapping were obtained using the Vertex-KMOS instrument mounted on the Very

* E-mail: e.vance@nimbus-astro.org (ERV)